Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112117
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Id : 691121201 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The ratio of mass percentage (w/w) of C : H in a hydrocarbon is 12 : 1. It has two carbon atoms. The weight (in g) of $CO_2(g)$ formed when 3.38 g of this hydrocarbon is completely burnt in oxygen is : (Given : Molar mass in g mol^{-1} C : 12, H : 1, , O : 16)

Options :

691121681. 5.68

691121682. 11.44

691121683. 22.74

691121684. 17.05

Question Number : 52 Question Id : 691121202 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The first and second ionization constants of a weak dibasic acid H_2A are 8.1×10^{-8} and 1.0×10^{-13} respectively. 0.1 mol of H_2A was dissolved in 1L of 0.1 M HCl solution. The concentration of HA^- in the resultant solution is :

Options :

691121685. 0.1 M

691121686. 9.53×10^{-6} M

691121687. 8.1×10^{-8} M

691121688. 1.0×10^{-13} M

Question Number : 53 Question Id : 691121203 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

SF_4 is isostructural with :

A. $BrF_4^{\ominus}$

B. CH_4

C. $IF_4^{\oplus}$

D. XeF_4

E. XeO_2F_2

Choose the **correct** answer from the options given below :

Options :

691121689. C Only

691121690. C and E Only

691121691. A and D Only

691121692. B and E Only

Question Number : 54 Question Id : 691121204 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Gas 'A' undergoes change from state 'X' to state 'Y'. In this process, the heat absorbed and work done by the gas is 10 J and 18 J respectively. Now gas is brought back to state 'X' by another process during which 6 J of heat is evolved. In the reverse process of 'Y' to 'X',

Options :

691121693. 18 J of the work is done by the gas 'A'.

691121694. 2 J of the work is done by the gas 'A'.

691121695. 12 J of the work is done on the gas 'A' by the surrounding.

691121696. 14 J of the work is done on the gas 'A' by the surrounding.

Question Number : 55 Question Id : 691121205 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Solution A is prepared by dissolving 1 g of a protein (molar mass = 50000 g mol^{-1}) in 0.5 L of water at 300 K. Its osmotic pressure is x bar. Solution B is made by dissolving 2 g of same protein in 1 L of water at 300 K. Osmotic pressure of solution B is y bar. Entire solution of A is mixed with entire solution of B at same temperature. The osmotic pressure of resultant solution is z bar. x, y and z respectively are :

(R = 0.083 L bar mol^{-1} K^{-1})

Options :

691121697. 9.96×10^{-4}; 9.96×10^{-4}; 9.96×10^{-4}

691121698. $9.96 \times 10^{-4}; 9.96 \times 10^{-4}; 19.92 \times 10^{-4}$

691121699. $4.98 \times 10^{-4}; 4.98 \times 10^{-4}; 9.96 \times 10^{-4}$

691121700. $4.98 \times 10^{-4}; 4.98 \times 10^{-4}; 4.98 \times 10^{-4}$

Question Number : 56 Question Id : 691121206 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

At 25°C, 20.0 mL of 0.2 M weak monoprotic acid HX is titrated against 0.2 M NaOH. The pH of the solution (a) at the start of the titration (when NaOH has not been added) and (b) when 10 mL of NaOH is added respectively, are :

Given : $K_a = 5 \times 10^{-4}$

$pK_a = 3.3$

$\alpha << 1$

(a) (b)

Options :

691121701. 0.7 2.0

691121702. 2.0 3.3

691121703. 1.1 2.2

691121704. 3.0 2.2

Question Number : 57 Question Id : 691121207 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

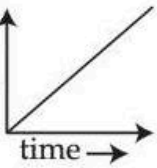
Consider the reaction $aX \rightarrow bY$, for which the rate constant at 30°C is $1 \times 10^{-3}\ mol^{-1}\ L\ s^{-1}$. Which of the following statements are true ?

A. When concentration of 'X' is increased to four times, the rate of reaction becomes 16 times.

B. The reaction is a second order reaction.

C. The half-life period is independent of the concentration of X.

D. Decomposition of N_2O_5 is an example of the above reaction.

E. $\ln\frac{[R_o]}{[R]}$ 

is valid for the above reaction.

Choose the **correct** answer from the options given below :

Options :

691121705. A and B Only

691121706. A, B and C Only

691121707. A, B, D and E Only

691121708. C and D Only

Question Number : 58 Question Id : 691121208 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The correct set that contains all kinds (basic, acidic, amphoteric and neutral) of oxides is :

Options :

691121709. Na_2O, K_2O, Al_2O_3 and As_2O_3

691121710. Al_2O_3, As_2O_3, CO and NO

691121711. K_2O, Cl_2O_7, As_2O_3 and NO

691121712. Na_2O, N_2O, Al_2O_3 and CO

Question Number : 59 Question Id : 691121209 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : The second ionization enthalpy of B, Al and Ga is in the order of B > Al > Ga.

Statement II : The correct order in terms of first ionization enthalpy is Si < Ge < Pb < Sn.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

691121713. Both **Statement I** and **Statement II** are true

691121714. Both **Statement I** and **Statement II** are false

691121715. **Statement I** is true but **Statement II** is false

691121716. **Statement I** is false but **Statement II** is true

Question Number : 60 Question Id : 691121210 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : Among Zn, Mn, Sc and Cu, the energy required to remove the third valence electron is highest for Zn and lowest for Sc.

Statement II : The correct order of the following complexes in terms of CFSE is $[Co(H_2O)_6]^{2+} < [Co(H_2O)_6]^{3+} < [Co(en)_3]^{3+}$.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

691121717. Both **Statement I** and **Statement II** are true

691121718. Both **Statement I** and **Statement II** are false

691121719. **Statement I** is true but **Statement II** is false

691121720. **Statement I** is false but **Statement II** is true

Question Number : 61 Question Id : 691121211 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following complexes will show coordination isomerism ?

A. $[Ag(NH_3)_2][Ag(CN)_2]$

B. $[Co(NH_3)_6][Cr(CN)_6]$

C. $[Co(NH_3)_6][Co(CN)_6]$

D. $[Fe(NH_3)_6][Co(CN)_6]$

E. $[Co(NH_3)_6][Fe(CN)_6]$

Choose the **correct** answer from the options given below :

Options :

691121721. B, C and D Only

691121722. B, D and E Only

691121723. A, C and D Only

691121724. C, D and E Only

Question Number : 62 Question Id : 691121212 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Complete combustion of X g of an organic compound gave 0.25 g of CO_2 and 0.12 g of H_2O. If the % of carbon is 25% and of hydrogen is 4.89%, then $X = ____ \times 10^{-3}$ g. (Nearest integer) (Molar mass of C, H and O are 12, 1 and 16 g mol^{-1} respectively.)

Options :

691121725. 273

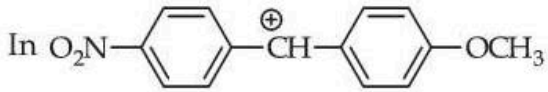
691121726. 27

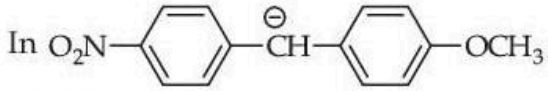
691121727. 2730

691121728. 227

Question Number : 63 Question Id : 691121213 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement I : In  , the carbocation is stabilised by +R effect of $-OCH_3$ group.

Statement II : In  , the carbanion is stabilised by −R effect of $-NO_2$ group.

In the light of the above statements, choose the **correct** answer from the options given below :

Options :

691121729. Both **Statement I** and **Statement II** are true

691121730. Both **Statement I** and **Statement II** are false

691121731. **Statement I** is true but **Statement II** is false

691121732. **Statement I** is false but **Statement II** is true

Question Number : 64 Question Id : 691121214 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The compound (X) on

(i) on heating in the presence of anhydrous $AlCl_3$ and HCl gas gives 2,4-dimethyl pentane

(ii) aromatization gives toulene and

(iii) cyclisation gives methyl cyclohexane

The correct name of compound (X) is :

Options :

691121733. Hept-2-ene

691121734. Hept-1,3,5-triene

691121735. Heptane

691121736. Hept-2,4,6-triene

Question Number : 65 Question Id : 691121215 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Correct statements regarding alkyl halides (R – X) among the following are :

A. Alcohol being less polar solvent as compared to water, alcoholic KOH favours elimination reaction with R – X.

B. Order of reactivity towards S_N1 mechanism is $C_6H_5-CH_2-Cl > C_6H_5-CHCl-C_6H_5$.

C. Non substituted aryl halides exhibit properties similar to alkyl halides.

D. Vinyl chloride is an example of haloalkene and allyl chloride is an example of haloalkyne.

E. R – Cl can be prepared by reacting R – OH with $SOCl_2$ but Ar – Cl cannot be prepared by reacting Ar – OH with $SOCl_2$.

Choose the **correct** answer from the options given below :

Options :

691121737. A, B and C Only

691121738. B and D Only

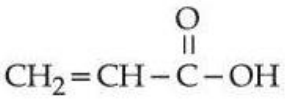
691121739. A and E Only

691121740. D and E Only

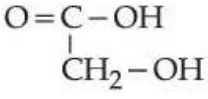
Question Number : 66 Question Id : 691121216 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

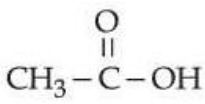
An organic compound "*x*" where molar ratio of C, O and H are equal, on treatment with 50% KOH under reflux followed by acidification produced "*y*". The most likely structure of "*y*" is :
[Molar mass of '*x*' is 58 g mol^{-1}]

Options :

691121741. 

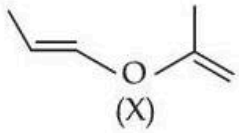
691121742. 

691121743. 

691121744. 

Question Number : 67 Question Id : 691121217 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A molecule (X) with following structure under mild acidic condition is hydrolysed to produce (Y) and (Z). Identify the correct statements about (Y) and (Z).



(X)

A. Both (Y) and (Z) have same molar mass.

B. (Y) and (Z) can be distinguished from each other by $NaHCO_3$.

C. (Y) and (Z) react with HCN with same rates.

D. (Y) and (Z) undergo addition reaction with 2,4-DNP.

Choose the **correct** answer from the options given below :

Options :

691121745. A, B and C Only

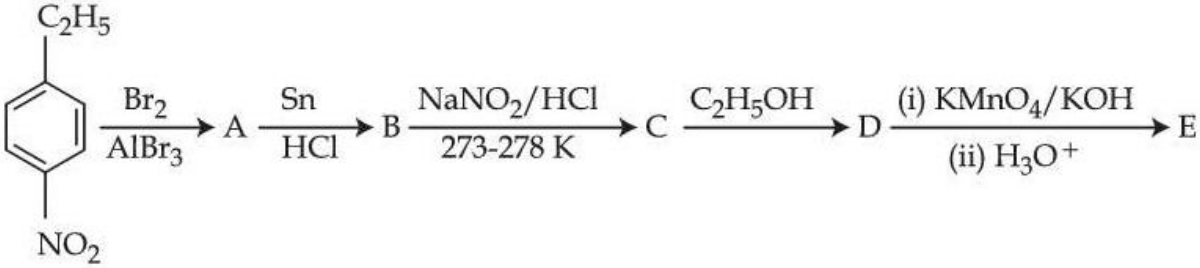
691121746. B and C Only

691121747. C and D Only

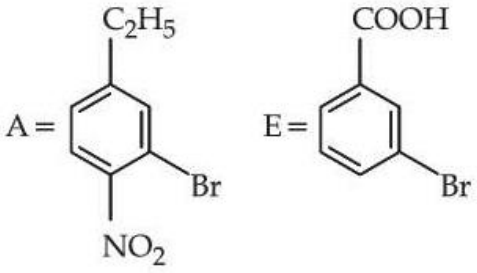
691121748. A and D Only

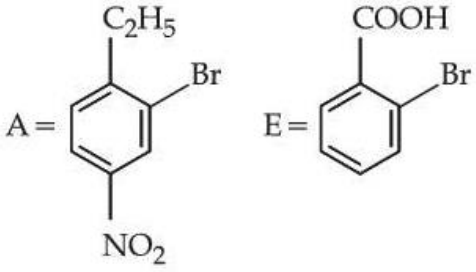
Question Number : 68 Question Id : 691121218 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

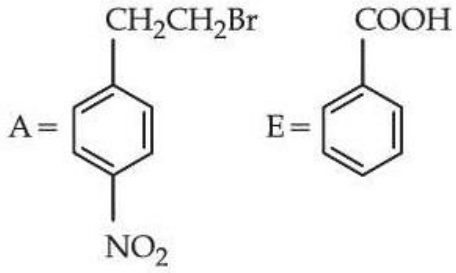
Identify compounds A and E in the following reaction sequence.

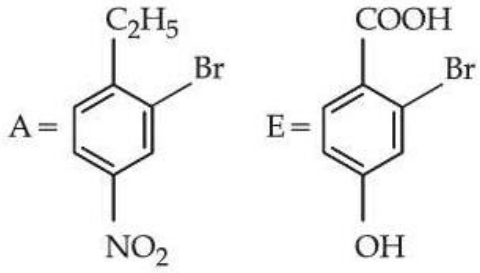


Options :

691121749. 

691121750. 

691121751. 

691121752. 

Question Number : 69 Question Id : 691121219 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Identify the correct pair having amino acid (A) and the hormone (B) that is iodinated derivative of the amino acid (A).

(T and Y represent one letter code for amino acids)

Amino acid (A)	Hormone (B)

Options :

691121753. T — Insulin

691121754. T — Thyroxine

691121755. Y — Thyroxine

691121756. Y — Insulin

Question Number : 70 Question Id : 691121220 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Among Fe^{2+}, Fe^{3+}, Cr^{2+} and Zn^{2+}, the ion that shows positive borax bead test and with highest ionisation enthalpy is :

Options :

691121757. Fe^{2+}

691121758. Zn^{2+}

691121759. Cr^{2+}

691121760. Fe^{3+}

Chemistry Section B

Section Id :	69112118
Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69112118
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 71 Question Id : 691121221 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The surface of sodium metal is irradiated with radiation of wavelength x nm. The kinetic energy of ejected electrons is 2.8×10^{-20} J. The work function of sodium is 2.3 eV. The value of x is _____ $\times 10^2$ nm. (Nearest integer)

(Given : $h = 6.6 \times 10^{-34}$ J s ; 1 eV $= 1.6 \times 10^{-19}$ J; $c = 3.0 \times 10^8$ m s^{-1})

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 72 Question Id : 691121222 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the following gas phase reaction being carried out in a closed vessel at 25°C.

$$2A(g) \longrightarrow 4B(g) + C(g)$$

$\frac{\text{time}}{(\text{min})}$	total pressure of the $\frac{\text{system}}{(\text{mm Hg})}$
30	300
∞	600

The pressure of C(g) at 30 minutes time interval would be _____ mm Hg. (nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 73 Question Id : 691121223 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the following two half-cell reactions along with the standard reduction potential given :

$$CO_2 + 6H^+ + 6e^- \longrightarrow CH_3OH + H_2O \qquad E^o_{red} = 0.02\ V$$

$$\frac{1}{2}O_2 + 2H^+ + 2e^- \longrightarrow H_2O \qquad E^o_{red} = 1.23\ V$$

A fuel cell was set up using the above two reactions such that the cell operates under the standard condition of 1 bar pressure and 298 K temperature. The fuel cell works with 80% efficiency. If the work derived from the cell using 1 mol of CH_3OH is used to compress an ideal gas isothermally against a constant pressure of 1 kPa, then the change in the volume of the gas,

$\Delta V =$ _____ m^3. (nearest integer)

Given : $F = 96500\ C\ mol^{-1}$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 74 Question Id : 691121224 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Number of paramagnetic ions among the following d- and f-block metal ions is _____.

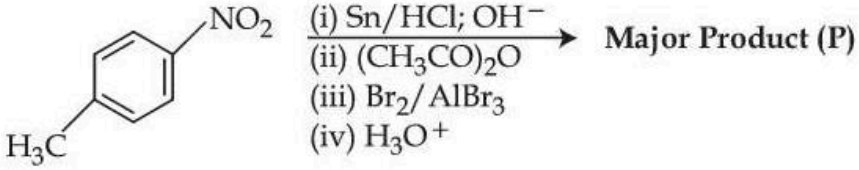
$Mn^{2+}, Cu^{2+}, Zn^{2+}, Yb^{2+}, Sc^{3+}, La^{3+}, Gd^{3+}, Lu^{3+}, Ti^{4+}, Ce^{4+}$

(Atomic number of Mn = 25, Cu = 29, Zn = 30, Yb = 70, Sc = 21, La = 57, Gd = 64, Lu = 71, Ti = 22, Ce = 58)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 75 Question Id : 691121225 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the following reactions sequence



When the product (P) is subjected to Carius analysis using $AgNO_3$, 1.0 g of the product (P) will produce _____ g of the precipitate of AgBr. (Nearest Integer)

(Given : molar mass in g mol^{-1} C : 12, H : 1, O : 16, N : 14, Br : 80, Ag : 108)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes